Expert Interview

Our Questions:

1. Expert Demography & Methodology

Years of Studies/Practice:

Specialization: Are they perfumers, high-sensitivity "expert smellers," or flavor chemists?

Examples:

Table 2: Experts on interactive olfactory experiences interviewed for Study 1. These experts covered a range of backgrounds and media for all ages (from children to adults) and audience scales (individual to theater experiences).

Expert	Background	Relevant projects or experience
E1	Filmmaker, olfactory artist	Olfactory experience designer for film screenings with scent, scented workshops and activities with children and people with disabilities, etc.
E2	Artist, exhibit and experience developer, playworker	Olfactory experience designer for film screenings with scent, olfactory experiences for children, smell experiences for neighborhood communities.
E3	Film programmer	Olfactory experience designer for film screenings with scent.
E4	Game designer, researcher	Academic researcher on scent in games, olfactory game designer.
E5	Game designer, programmer	Lead designer for commercially distributed video game with scratch-and-sniff card (over 750,000 copies sold).
E6	Game designer, programmer	Lead designer for commercially distributed text-adventure with scratch-and-sniff card (over 130,000 copies sold).
E7	Game designer, researcher	Interactive art installations and games exhibited internationally, sensory design & teaching.
E8	Artist, HCI researcher	Interactive olfactory installations, research on olfactory HCI.

Table 1: Demographics of Participants in the Formative Study. 'Exp.' denotes their user experience within their respective domains. 'Task' indicates the specific tasks performed according to their provided dialog history. 'Rounds' represents the number of conversation rounds conducted.

ID	Domain	Exp.	Task	Interaction Mode/LLM Name	Rounds
P1	Design (Ph.D.)	1.5 Years	Web article crawlers with python	Coding Mode/Qwen	11
P2	Electric Grid (UG)	0.5 Years	Signal analysis with Matlab	Chatting Mode/Doubao	13
P3	Clinical Medicine (Ph.D.)	6 Years	Medical image processing and classification with Python	Chatting Mode/Kimi	21
P4	Theory Math (Ph.D.)	3 Years	Table drawing and adjusting with Latex	Coding Mode/Qwen	8
P5	Policy Making (M.S.)	1 Year	Interview data analysis and visualization with python	Coding Mode/GPT-4o	13
P6	Economics (M.S.)	5 Years	Financial data analysis and visualization with python	Data Analysis Mode/GPT-4o	29

2. Temporal & Perceptual Justification

- What are the differences between mixture and sequence? Can we use sequence to create mixtures of smell?
  Expected Answer: Sequence is fine.
- Is using essential oils for mixtures a common practice in the fragrance industry?
  Expected Answer: Essential oil is fine.
- What is the best duration to let users perceive smell? (Tell them "Our users find 10–15 seconds "too fast" to reflect on the scent")
  Expected Answer: 60s is fine.
- How many active ingredients do we need for the mixture/sequence? To reach the "sweet spot" for complexity without being "confusing"?
  Expected Answer: 3 to 6 if fine. Explain why having too many (e.g., 10+) or too few fails to replicate a realistic food smell.
- What is volatility of smell? Does it affect the order of sequence? (In the Methods section our LLM decision is affected by volatility of smell.) Is there a professional "top-middle-base note" logic or a "buildup" strategy that experts use?

Expected Answer: The relationship between volatility and buildup strategy etc. and It is fine to show LLM the volatility

3. Scent Space & Hardware Constraints

- How many base odors do we need?
  Expected Answer: 12 is fine. (Or: the more the better, if 12 is hardware limit 12 is fine.)
- What choices of base odors do you recommend? (Show them the Formative Study word frequency.) Give them our choice of base odors. Do you think our choice is good enough?
  Expected Answer: Yes, our choice of base odors is good enough.
  Important to let reviewers know our Base Odors are backed up by experts
- Can you give us the volatility and notes of our chosen 12 base smells? (Or we give them they confirm)
  Show them our scent_classification.json
  Expected Answer: The volatility and notes of the 12 base smells
  Important to let reviewers know the Volatility and Notes of base odors are backed up by experts.
  If the perfumers cannot smell our essential oils in person, they can give us volatility + notes from their past experiences

4. Few Shot Examples

- Can you give us 3 example of the mixtures? Example:

> User: "The burger smells savory like cheese, it has a lot of lettuce and ketchup in it so somehow smells fresh" (Let the expert decide both user input and assistant output)
> Assistant:
> ```
> {
> "Cumin": 0.20,
> "Eucalipto": 0.00,
> "Onion": 0.25,
> "ylang ylang": 0.00,
> "Sichuan oil": 0.00,
> "Rec clover": 0.15,
> "Cypress": 0.00,
> "Thyme": 0.10,
> "Strawberry": 0.00,
> "Isoveric Acid": 0.20,
> "Cinnamon": 0.00,
> "Sage": 0.10
> }
> ```
> The sum of ratio is 1, and then we map to 60s in sequence.